YIHENG LYU

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Github ⋄ LinkedIn

EDUCATION

University of Stuttgart, Germany *Starting October 2026*
M.Sc. in Electrical Engineering **Nanjing Tech University, China** *June 2026*
B.Eng. in Automation
GPA: 88/100
Relevant course: Classic Control Theory, Modern Control Theory, Discrete Control, Robotic Control, Machine Learning, Digital Signal Processing, Embedded Systems

RESEARCH INTERESTS

I am interested in Robotic Manipulation, Adaptive Control, Nonlinear Control Systems, Learning-based Control, Intelligent Autonomous Systems, Distributed Optimization.

RESEARCH EXPERIENCE

Franka Gazebo Adaptive Control with Model Mismatch and Disturbance
May 2025 - Present Independent Research Project

- *github.com/hengsleep/Franka_Gazebo_RBF_adaptive_control_Prj*
- Developed a ROS 2 and Gazebo-based simulation framework for adaptive torque control of a 7-DOF Franka FR3 robotic manipulator under model uncertainty and external disturbances.
- Integrated computed torque control with online radial basis function (RBF) neural network compensation to improve trajectory tracking performance under uncertain robot dynamics.
- Implemented robot dynamics modeling using Pinocchio and developed a custom torque controller based on `ros2_control` with real-time execution in C++.
- Validated the proposed framework in Gazebo Sim with physical parameter mismatch and disturbance injection scenarios.

Adaptive Control of Robotic Manipulators with RBF and σ-Modification
Jan 2026 - Jun 2026 Bachelor Thesis Research

- *github.com/hengsleep/RBF_Sigma_Adaptive_Robot_control_Prj*
- Proposed a nonlinear adaptive control framework for uncertain robotic manipulators using radial basis function (RBF) neural networks for online approximation of unknown dynamics.
- Developed a computed torque controller combined with Lyapunov-based adaptive laws and σ-modification to improve robustness against model uncertainties.
- Established uniformly ultimately bounded (UUB) tracking performance through Lyapunov stability analysis.
- Evaluated the proposed controller through MATLAB/Simulink simulations under unknown dynamics and external disturbances.

Distributed Adaptive Resource Allocation for Multi-Agent Systems
Jan 2025 - Nov 2025 Research Project

- *github.com/hengsleep/DARA_Lipschitz_continuous_Prj*
- Investigated distributed resource allocation problems for multi-agent systems over directed communication networks with limited information exchange.
- Developed distributed adaptive optimization algorithms to achieve cooperative decision-making among heterogeneous agents with locally available objective functions.

- Analyzed convergence properties of distributed algorithms under heterogeneous Lipschitz continuous cost functions and network topology constraints.
- Evaluated the proposed algorithms through numerical simulations on large-scale networked multi-agent systems.

INTERNSHIP EXPERIENCE

Automation Engineering Intern – 3M Solventum
Jun 2024 - Sep 2024 Shanghai, China

- Received training in Rockwell RSLogix 5000 PLC programming, including ladder logic development and industrial automation control workflows.
- Participated in a machine vision-based quality inspection project, involving industrial camera data acquisition and image-based defect detection for manufacturing processes.
- Gained hands-on exposure to automated manufacturing systems, including KUKA robotic manipulators, pneumatic conveying systems, motion control systems, and process monitoring equipment.
- Learned industrial communication and control architectures, including Cisco Converged Plantwide Ethernet (CPwE), Process Control Network (PCN), and SCADA-based monitoring systems.

THESES AND RESEARCH REPORTS

1. *Adaptive Control of Multi-joint Robotic Manipulator Systems with RBF Neural Networks*
Bachelor Thesis, 2026.
2. *Distributed Adaptive Resource Allocation under Lipschitz Continuous Cost Functions*
Research Lab Thesis, 2025.

LANGUAGE SKILLS

1. English: C1 (IELTS 7.0)
2. German: A1 (Basic proficiency)

ACHIEVEMENTS

Academic Scholarship, Nanjing Tech University 2022–2026
Awarded annually for academic excellence, ranking among the top 3%, 5%, and 10% of students.
MCM (Mathematical Contest in Modeling), COMAP, USA 2024
Honorable Mention.
RAICOM Robotics Competition, RAICOM International Organizing Committee 2024
Provincial Second Prize.
A-ST Network Technology Competition, Computer Education Research Association 2023
Provincial Third Prize.

TECHNICAL SKILLS

Programming Languages	C/C++, Python, MATLAB, LaTeX
Robotics Frameworks	ROS 2, Gazebo Sim, MATLAB/Simulink
Control and Simulation	Adaptive Control, Nonlinear Control, Optimization
Machine Learning	PyTorch, TensorFlow
Engineering Tools	Git, Linux, PLC (Siemens & Rockwell)